ARQUITECTURAS E INFRAESTRUCTURAS PARA INTELIGENCIA ARTIFICIAL
PRACTICE 1: CLOUD COMPUTING SERVICES

Objectives

- Learn how to offload AI processing to the cloud to observe the advantages of Cloud Computing.

Introducción:

The exercise consists of developing a Smart City application to control an **Intelligent Lighting System** (SLS – Smart Lighting System) in a specific area of the city. An intelligent lighting system is responsible for managing the switching on/off of luminaires based on the presence of sensitive elements (people or vehicles) within a defined control zone.

A growingly common approach involves using digital cameras as presence detection sensors.

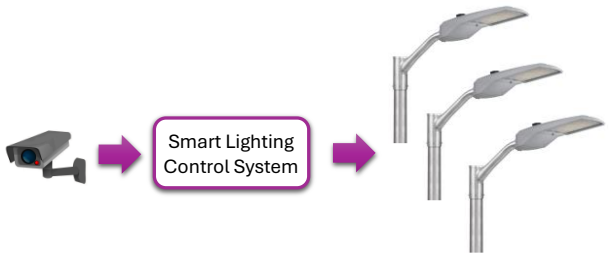


Fig. 1: System overview

In this exercise, public webcams will be used to capture images of the control zones, and a local machine will serve as the processor for the digital camera. As you know, the internet offers a large number of publicly accessible webcams that stream continuous video. For example, the following:

http://www.insecam.org/
https://www.earthcam.com/
https://www.skylinewebcams.com/
https://www.webcamtaxi.com/en/

This video stream will be processed to identify the elements that determine how the controller operates the lighting system.

PART 1: LOCAL WORK

Task 1.1. Identify a public webcam that shows a street where people or vehicles are circulating. The camera must be fixed, for example, like the one shown in the following figure:

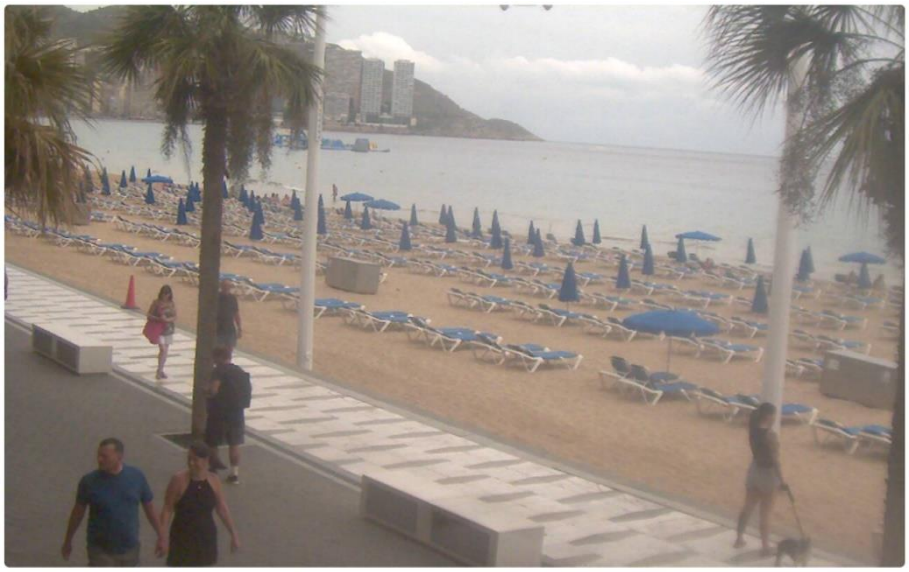


Fig. 2: Webcam example.

Task 1.2. On a local machine (laptop or lab computer), create a process that captures a video frame from the webcam every 5 seconds. The frame resolution must be sufficient to identify the elements.

Note: Use the sample process provided by the Teaching Team.

Task 1.3. On a local machine (laptop or lab computer), create a process to count the number of people in each frame.

Use the Ultralytics YOLOv8 AI model (https://ultralytics.com/yolov8), which is capable of performing a wide range of automatic tasks such as object tracking, instance segmentation, image classification, and pose estimation.

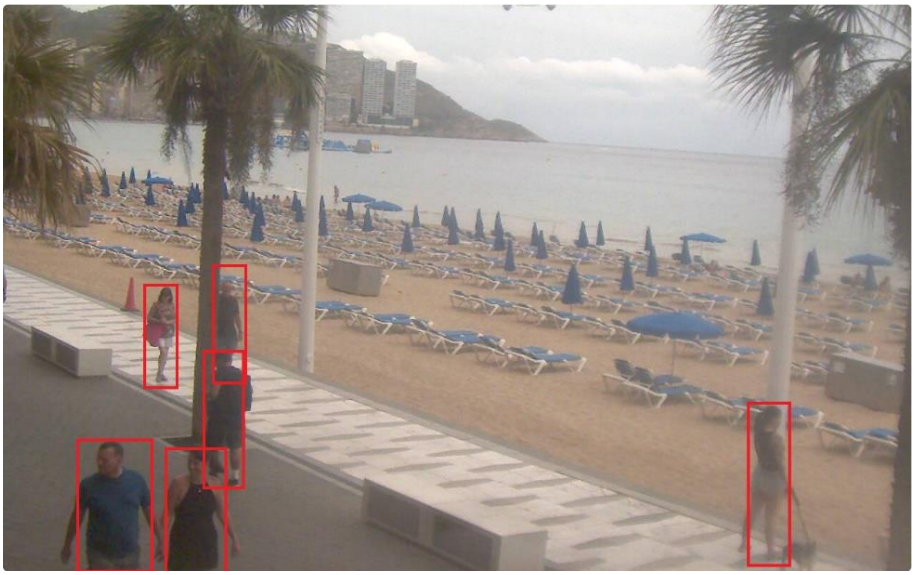


Fig. 3: Person detection

Save the number of people along with the date in a CSV file and draw a graph that illustrates the number of people per frame over time.

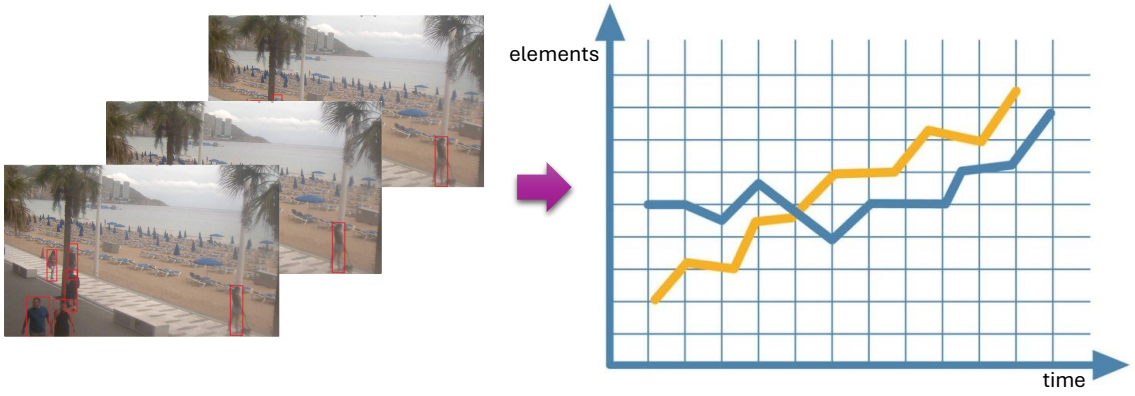


Fig. 4: Elements detection evolution

PART 2: OUTSOURCING PROCESSING TO CLOUD VIRTUAL MACHINES

In this second part, the element detection processing will be outsourced to the cloud.



Fig. 5: Outsourcing overview

Task 2.1. Create a virtual machine (VM) in the Microsoft Azure cloud (it is recommended to use the "Standard B1s" size). Install the necessary library packages to implement the services for this practice: python3-opencv, python3-pip.

Note 1: You can refer to the example documentation provided by the Teaching Team.

Note 2: If an error such as "killed" appears during package installation, it is due to RAM exhaustion and the system terminating the process. In that case, add the --no-cache-dir option.

Task 2.2. On your local machine, capture a frame from the webcam using the process developed in Task 1.2, and send it to the cloud VM for processing.

Note: Use the example process provided by the Teaching Team to send the data.

Task 2.3. On the VM created in the cloud, perform a process to count the number of people in each frame. Use the Ultralytics YOLOv8 AI model used in Task 1.3. Store the results in a CSV file.

Task 2.4. Show a histogram and other data on a web page hosted on the cloud VM. The page should be accessible using a public IP address.

Task 2.5. Obtain performance data from the execution of your cloud VM: detection efficiency, memory usage, CPU usage, bandwidth consumption, monetary cost, response time.

PARTE 3: EXTERNALIZACIÓN DEL PROCESAMIENTO BAJO EL MODELO SERVERLESS EN LA NUBE

A serverless application will be created to run frame processing on MS Azure.



Fig. 6: Outsourcing overview with Azure Functions

Task 3.1. Create a static web page in an Azure storage account. This web page will be used to display the data we previously generated.

Task 3.2. On your local machine, capture a frame from the webcam using the process developed in Task 1.2, and send it to a cloud storage container different from the one used for the web page, for processing. This process should run continuously, for example, every 0.5 seconds.

Note: Use the sample process provided by the Teaching Team to send the data.

Task 3.3. Create a serverless application to run on Microsoft Azure. You must create a blob trigger function so that it executes every time a raw image is uploaded to the container.

WARNING: Make sure the function only runs when an image is uploaded to the raw images container. Also ensure that the container where you upload raw images is different from the container where the web page files are stored. A misconfiguration here could cause the function to also trigger when it writes the processed image to the web page container. The result is a recursive execution of the function, which could quickly consume your Azure credits due to the instantiation of a large number of functions.

Microsoft Azure offers certain types of services related to high-performance computing. These services provide GPU infrastructure and high-performance computing in clusters with large amounts of resources.

They also offer AI processing services such as **Azure Machine Learning**. This service allows you to allocate the necessary computing resources and pay per use. It provides access to simple CPU and GPU resources, as well as large compute clusters with many nodes and memory:

https://azure.microsoft.com/es-es/products/machine-learning#tabx69c4fee3c8844375955db36c4f74da17

The AI services available in Azure consist of APIs that allow you to integrate these services into applications:

https://learn.microsoft.com/es-es/azure/ai-services/

In this part, we will replace the YOLOv8 AI model used in previous sections with an Azure AI service. This will be done using the following **Computer Vision AI service for image tagging**:

https://learn.microsoft.com/es-es/azure/ai-services/computer-vision/concept-tag-images-40

This service allows you to tag the objects that appear in the image.

Task 3.4. Create an AI Service that performs the required processing in Microsoft Azure.
Note: You can base this on the sample documentation provided by the Teaching Team.

Task 3.5. Develop a serverless function that processes the raw images uploaded to the storage container, using the Azure AI service to count the number of people present in the image. Draw a box around each detected person using the "bounding_box" property provided by the AI service.

Task 3.6. Obtain performance data from the cloud application execution: detection efficiency, monetary cost, response time.

PART 4: MULTI-CAMERA MANAGEMENT

In an SLS, the control system manages multiple connected cameras and a collection of luminaires that fully cover the area to be monitored.



Fig. 5: Wide control area

The performance of local processing is highly dependent on the resources of the available machine, whether on the IoT sensor devices themselves (digital cameras) or on low-cost Edge devices (Arduino, Raspberry Pi).

Task 4.1. Identify an increasing number of webcams from which frames can be sent for analysis. These frames may be sent from the same local device or from multiple devices at a rate of 1 frame every 5 seconds per camera.

Task 4.2. Send frames from new cameras to the VM created in Part 2 and obtain performance data to identify the processing limit—that is, how many simultaneous cameras it can handle. Obtain performance data from the execution of your cloud VM: memory usage, CPU usage, bandwidth consumption, monetary cost, response time.

Task 4.3. Send frames from new cameras to the Cloud application created in Part 3 and obtain performance data to identify the processing limit—that is, how many simultaneous cameras it can handle. Obtain performance data from the execution of the cloud application: detection efficiency, monetary cost, response time.

For all parts:
Write a project report describing the steps followed in each part, the problems encountered, and the design decisions made to complete the required work. Add a section with conclusions about the work carried out. Include any screenshots you consider necessary to demonstrate that you followed the steps indicated in the instructions.

Advanced (+2 points)

1. Based on the developed system, create an intelligent system that counts the number of people passing by each camera over a period of time. The passage of the same person across multiple frames should be counted as a single individual.
2. Based on the developed system, create an intelligent system that counts both people and vehicles passing by each camera over a period of time.
3. Design and develop improvements to optimize the performance of the architectures developed in each part. For example:
   - Set a minimum image resolution that maintains detection efficiency.
   - Compress images to reduce bandwidth usage.
   - Implement motion detection techniques on the local machine.
   - etc.

Submission guidelines:

- The project can be completed in groups of 2 or 3 students.
- Submission must be done through the Moodle assignment.
- The accepted document format is ".pdf".